

Surface area



Key idea

Surface area is the total area of all the outside faces of a 3D shape. For a net, you find the area of each shape in the net and add them together.

Steps

1. Identify each face of the 3D shape.
2. Find the area of each face.
3. Add all the areas together.
4. Write the answer with square units (cm^2 , m^2 , etc.).

Common mistake

Forgetting to include both ends. A cylinder has two circular faces (top and bottom), not one. A cuboid has 6 faces, not 5.

Cube

A cube has side length 4 cm.
Each face is a square: $4 \times 4 = 16 \text{ cm}^2$.
A cube has 6 faces.
 $6 \times 16 = 96 \text{ cm}^2$.
Answer: **96 cm^2**

Cuboid

A box is 8 cm long, 3 cm wide, 2 cm high.
Top & bottom: $2 \times (8 \times 3) = 48 \text{ cm}^2$
Sides: $2 \times (8 \times 2) = 32 \text{ cm}^2$
Ends: $2 \times (3 \times 2) = 12 \text{ cm}^2$
Total: $48 + 32 + 12 = 92 \text{ cm}^2$.
Answer: **92 cm^2**

Try these

1. A cube has side length 6 cm. Find its surface area.
2. A rectangular prism is 10 cm by 4 cm by 2 cm. Find its surface area.